

Estimation & Hypothesis Testing

Probability & Statistics

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1. What is Estimation?

From sample to population

Definition

Estimation is the process of using sample data to infer the value of an unknown **population parameter**.

Why do we need it?

- Measuring entire populations is **impossible** or **prohibitively expensive**
- We collect a **sample** → compute a **statistic** → estimate the **parameter**
- Accounts for sampling variability and uncertainty

Examples:

- Estimate **true average API latency** from 1000 logged requests
- Estimate **proportion of users** experiencing a specific bug
- Estimate **mean server uptime** between failures
- Estimate **true error rate** of a machine learning classifier

Key Idea

Estimation answers: *"Based on what we observed, what is the most reasonable guess for the whole population?"*

2. Point Estimators

Single-value estimates

Definition

A **point estimator** is a statistic $\hat{\theta}$ calculated from sample data that provides a **single best guess** for a population parameter θ .

Parameter	Symbol	Point Estimator
Population mean	μ	$\hat{\mu} = \bar{X} = \frac{1}{n} \sum X_i$
Population variance	σ^2	$\hat{\sigma}^2 = S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
Population proportion	p	$\hat{p} = \frac{X}{n}$ (successes / total)

IT Context

If you measure response times of 50 API calls and get $\bar{x} = 142$ ms, then **142 ms** is your point estimate for the true population mean latency μ .

Properties of Good Point Estimators

What makes an estimator reliable?

Three Key Properties:

- 1 **Unbiasedness:** $E[\hat{\theta}] = \theta$
On average, hits the true value
- 2 **Consistency:** $\hat{\theta} \rightarrow \theta$ as $n \rightarrow \infty$
Improves with more data
- 3 **Efficiency:** Smallest variance among unbiased estimators
Most precise for a given sample size

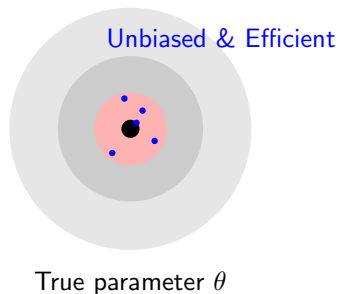


Figure: Bullseye analogy for estimator quality

Note

$\bar{X}$ and S^2 are unbiased, consistent, and efficient for normal populations. $\hat{p}$ is unbiased for any population.

3. Interval Estimation

Confidence Intervals (CI)

Definition

A **confidence interval** provides a **range of plausible values** for a population parameter, along with a confidence level (e.g., 95%).

Formula (Mean, σ known or large n):

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Components:

- $\bar{x}$: Sample mean
- $z_{\alpha/2}$: Critical value from standard normal (1.96 for 95%)
- $\frac{\sigma}{\sqrt{n}}$: Standard error
- Margin of Error $E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Interpretation:

- **Correct:** “If we repeated this sampling 100 times, 95 intervals would contain μ .”
- **Wrong:** “There is a 95% probability μ is in this specific interval.” (Parameter is fixed!)

Practice

Interval Estimation: When σ is Unknown

Using the t-distribution

In practice, σ is rarely known. We replace it with sample standard deviation s and use the **Student's t-distribution**:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

Key Differences from z :

- t-distribution has **heavier tails** (accounts for extra uncertainty)
- Critical value depends on **degrees of freedom** $df = n - 1$
- As $n \rightarrow \infty$, $t \rightarrow z$

IT Example:

- $n = 25$ server logs, $\bar{x} = 165$ ms, $s = 22$ ms
- 95% CI: $165 \pm 2.064 \cdot \frac{22}{\sqrt{25}} = 165 \pm 9.1$
- Interval: **[155.9, 174.1] ms**

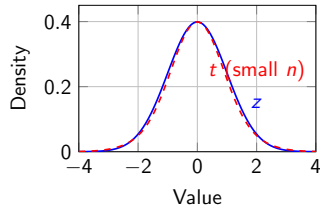


Figure: t vs z distribution

4. Choosing the Sample Size

Balance precision, cost, and time

To achieve a desired margin of error E at confidence level $(1 - \alpha)$:

For estimating a mean:

$$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

For estimating a proportion:

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 \hat{p}(1 - \hat{p})$$

If $\hat{p}$ unknown, use $\hat{p} = 0.5$ (most conservative, maximizes n)

Example: Server Latency Study

- Want 95% CI with ± 3 ms precision
- Pilot data suggests $\sigma \approx 15$ ms
- $n = \left(\frac{1.96 \times 15}{3} \right)^2 = (9.8)^2 \approx 96.04$
- **Always round up:** $n = 97$ requests

Factors Affecting Sample Size

What drives the number up or down?

Increases n when:

- Higher confidence level (99% $\hat{}$ 95%)
- Smaller margin of error (± 1 vs ± 5)
- Higher population variability ($\sigma \uparrow$)
- Proportion near 0.5 (maximum uncertainty)

Practical IT Trade-offs:

- More data $\rightarrow$ better precision but higher storage, compute, and collection cost
- Use pilot studies or historical logs to estimate σ or $\hat{p}$
- Consider sequential sampling: collect data, check width, stop when met

Rule of Thumb

- $n \geq 30$: Central Limit Theorem applies (approx. normal sampling distribution)
- $n < 30$: Use t-distribution, check normality assumption

Chapter 4: Hypothesis Testing

Definitions

① **Statistical hypothesis** A statistical hypothesis is a statement about a population parameter.

② **Null hypothesis and Alternative hypothesis**

The two complementary hypotheses in a hypothesis testing problem are called the null hypothesis and the alternative hypothesis. They are denoted by H_0 and H_1 , respectively.

Goal of Hypothesis testing

The goal of a hypothesis test is to decide, based on a sample from the population, which of two complementary hypotheses is true.

Tests of hypotheses for a simple sample

Hypothesis testing procedure

A hypothesis testing procedure or Hypothesis test is a rule that specifies

- 1 For which sample values the decision is made to accept H_0 as true.
- 2 For which sample values H_0 is rejected and H_1 is accepted as true.

Test Errors

The truth or falsity of a hypothesis can never be known with certainty unless the entire population is examined accurately and thoroughly.

Test Errors

Therefore, a hypothesis test based on a random sample may lead to one of the two types of wrong conclusions (Table 1):

- 1 Type I error: Reject H_0 when H_0 is true.
- 2 Type II error: Fail to reject H_0 when H_0 is false.

Table1: Decisions in hypothesis testing.

		Decision	
		Accept H_0	Reject H_0
Truth	H_0 is true	Correct decision	Type I error
	H_0 is false	Type II error	Correct decision

Probability of making error

- The probability of making a Type I error is denoted by α . That is,

$$P(\text{Type I error}) = P(\text{reject } H_0 | H_0 \text{ is True})$$

- The probability of making a Type II error is denoted by β . That is,

$$P(\text{Type II error}) = P(\text{fail to reject } H_0 | H_0 \text{ is false})$$

The Type I error is called the significance level, or the α , or the size of the test.

Null Hypothesis and Alternative Hypothesis

- The null hypothesis, denoted by H_0 , is typically a clear statement of equality. The unknown population parameter θ is equal to some specific constant value θ_0 . This hypothesis is written

$$H_0 : \theta = \theta_0.$$

- The alternative hypothesis, denoted by H_1 , is an assumption about θ that differs from θ_0 . The alternative hypothesis occurs in one of three general forms.

Alternative Hypothesis

Two sided alternative

It deals with possible differences from θ_0 on either side of θ_0 . This hypothesis is written as

$$H_1 : \theta \neq \theta_0$$

Left directional alternative

This alternative hypothesis states that θ is less than θ_0 . It is written as

$$H_1 : \theta < \theta_0$$

Right directional alternative

This alternative hypothesis states that θ is greater than θ_0 . It is written as

$$H_1 : \theta > \theta_0$$

General Procedure for Hypothesis Tests

Use of the following sequence of steps in applying hypothesis-testing methodology is recommended.

- ➊ From the problem context, identify the parameter of interest, θ .
- ➋ State the null hypothesis, H_0 .
- ➌ Specify an appropriate alternative hypothesis, H_1 .
- ➍ Choose a significance level, α .
- ➎ Determine an appropriate test statistic.
- ➏ State the rejection region for the statistic.
- ➐ Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
- ➑ Decide whether or not H_0 should be rejected and report that in the problem context.

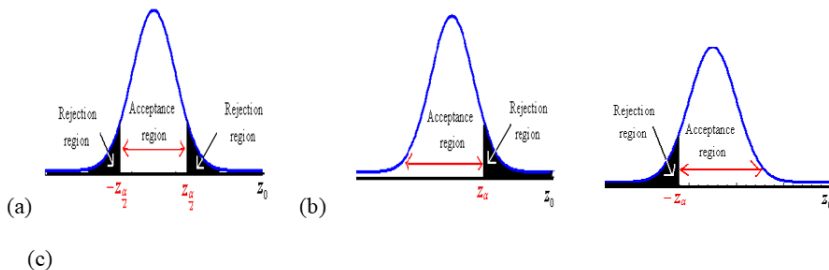
Tests about the Mean μ of a Normal Distribution: Variance σ^2 known: Distribution of Z_0

Fig.1 shows the distribution of the statistic z_0 when $H_0 : \mu = \mu_0$ is true for

(a) The two-sided alternative $H_1 : \mu \neq \mu_0$

(b) The one-sided alternative $H_1 : \mu > \mu_0$

(c) The one-sided alternative $H_1 : \mu < \mu_0$



Test procedure (Z test)

- **Step 1:** State H_0 and H_1 .

- ① $H_0 : \mu = \mu_0$.

- ② $H_1 : \mu \neq \mu_0$ for two sided test;

- $H_1 : \mu > \mu_0$ for upper-sided test;

- $H_1 : \mu < \mu_0$ for lower-sided test.

- **Step 2:** Determine a test statistics and its value

$$Z_0 = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}.$$

- **Step 3:** Determine a critical value of α .

- $Z_{\frac{\alpha}{2}}$ for two-sided test; Z_{α} for one-sided test.

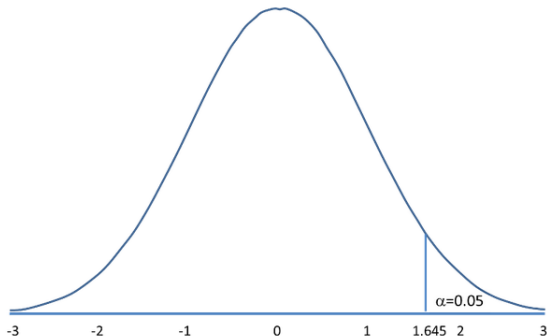
- **Step 4:** Make a conclusion. Reject H_0 if

- $|Z_0| > Z_{\frac{\alpha}{2}}$ for two-sided test.

- $Z_0 > Z_{\alpha}$ for upper-sided test;

- $Z_0 < -Z_{\alpha}$ for lower-sided test.

Upper-sided test



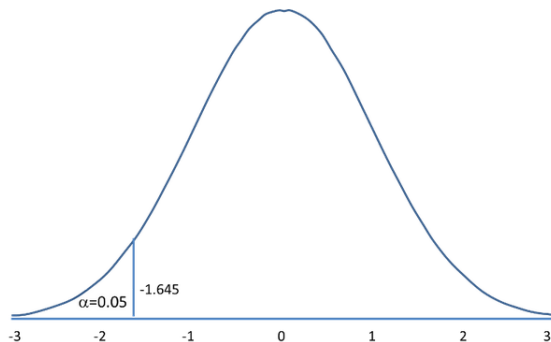
Rejection Region for Upper-Tailed Z Test ($H_1: \mu > \mu_0$) with $\alpha=0.05$

The decision rule is: Reject H_0 if $Z \geq 1.645$.

Upper-Tailed Test

α	Z
0.10	1.282
0.05	1.645
0.025	1.960
0.010	2.326
0.005	2.576
0.001	3.090
0.0001	3.719

Lower-sided test



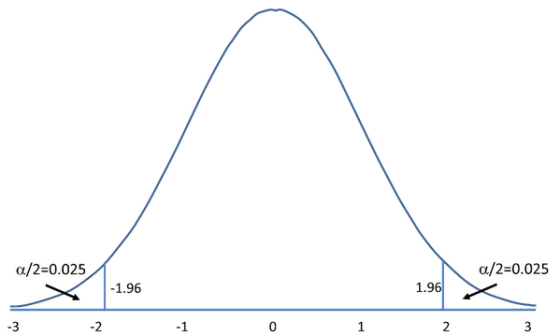
Rejection Region for Lower-Tailed Z Test ($H_1: \mu < \mu_0$) with $\alpha = 0.05$

The decision rule is: Reject H_0 if $Z \leq -1.645$.

Lower-Tailed Test

α	Z
0.10	-1.282
0.05	-1.645
0.025	-1.960
0.010	-2.326
0.005	-2.576
0.001	-3.090
0.0001	-3.719

Two-sided test



Rejection Region for Two-Tailed Z Test ($H_1: \mu \neq \mu_0$) with $\alpha = 0.05$

The decision rule is: Reject H_0 if $Z \leq -1.960$ or if $Z \geq 1.960$.

Two-Tailed Test	
α	Z
0.20	1.282
0.10	1.645
0.05	1.960
0.010	2.576
0.001	3.291
0.0001	3.819

Example

The mean lifetime of a sample of 100 fluorescent light bulbs produced by a company is computed to be 1570 hours with a standard deviation of 120 hours. If μ is the mean lifetime of all the bulbs produced by the company, test the hypothesis $\mu = 1600$ hours against the alternative hypothesis $\mu \neq 1600$ hours, using a level of significance of (a) $\alpha = 0.05$ and (b) $\alpha = 0.01$

Solution:

- We must decide between 2 hypotheses: $H_0 : \mu = 1600$ hours Vs $H_1 \neq 1600$ hours

$$Z_0 = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{1570 - 1600}{\frac{120}{\sqrt{100}}} = -2.5$$

- Critical values: a) $\alpha = 0.05$; $Z_{\frac{\alpha}{2}} = 1.96$; b) $\alpha = 0.01$; $Z_{\frac{\alpha}{2}} = 2.576$.

- we reject H_0 if $|Z_0| > Z_{\frac{\alpha}{2}}$

a) $|-2.5| > 1.96$, Thus we reject H_0 at 0.05 level of significance

b) $|-2.5| < 2.58$; Thus we accept H_0 at 0.01 level of significance.

Variance σ^2 unknown

Fig. 2 shows the distribution of the statistic t_0 when $H_0 : \mu = \mu_0$ is true for

- (a) The two-sided alternative $H_1 : \mu \neq \mu_0$
- (b) The one-sided alternative $H_1 : \mu > \mu_0$
- (c) The one-sided alternative $H_1 : \mu < \mu_0$

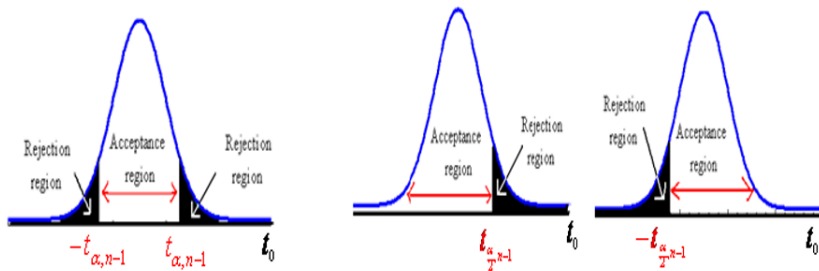


Fig.2: Distribution of t_0

Test procedure (t test)

- **Step 1:** State H_0 and H_1 .

- ① $H_0 : \mu = \mu_0$.

- ② $H_1 : \mu \neq \mu_0$ for two sided test;
 $H_1 : \mu > \mu_0$ for upper-sided test;
 $H_1 : \mu < \mu_0$ for lower-sided test.

- **Step 2:** Determine a test statistics and its value

$$t_0 = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}.$$

- **Step 3:** Determine a critical value of α .

$t_{\frac{\alpha}{2}, n-1}$ for two-sided test; $t_{\alpha, n-1}$ for one-sided test.

- **Step 4:** Make a conclusion. Reject H_0 if

$|t_0| > t_{\frac{\alpha}{2}, n-1}$ for two-sided test.

$t_0 > t_{\alpha, n-1}$ for upper-sided test;

$t_0 < -t_{\alpha, n-1}$ for lower-sided test.

Example

A sample ($n = 20, \bar{X} = 4.0, s = 0.83$) is taken from a normally distributed population that has a unknown μ and unknown σ . Test the hypothesis $H_0 : \mu = 3.6$ versus $H_1 : \mu \neq 3.6$ at $\alpha = 0.05$.

Solution:

- 1 Parameter of interest: Mean μ .
- 2 Hypotheses: $H_0 : \mu = 3.6$ versus $H_1 : \mu \neq 3.6$
- 3 Significance level: $\alpha = 0.05$
- 4 As the population is normally distributed and σ is unknown, we use the t statistics test. We have

$$t_0 = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{4.0 - 3.6}{\frac{0.83}{\sqrt{20}}} = 2.1552$$

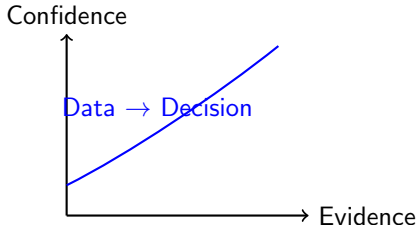
- 5 Critical value:

$$t_{\frac{\alpha}{2}, n-1} = t_{\frac{0.05}{2}, 20-1} = t_{0.025, 19} = 2.433$$

- 6 As $|2.1552| < 2.433$, we do not reject H_0 at the 0.05 level of significance.

Thank You!

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"All models are wrong, but some are useful."

— George E. P. Box